ALGEBRAIC CURVES

In this lecture, we present basic facts about algebraic curves. As before, We will not prove all the things covered here as this and the next few lectures are to be taken more like a glossary of terms and results that we will use. The students are expected to work out the details on their own, as this will give them much needed experience with these objects.

1. Curves and uniformizers

A **curve** will always mean **a projective variety of dimension** 1. We will typically use the following notations, some of which have already been introduced. K and $\bar{K}$ and G_K will be as before. C will denote a curve over $\bar{K}$ and $P \in C$ a point. If C is defined over K, we will say C/K at times and if P is also defined over K, we will say $P \in C(K)$. $\bar{K}(C)$ will denote the function field of C over $\bar{K}$ and similarly $K(C)$ will denote the function field of C over K. $\mathcal{O}_{C,P}$ denotes the local ring of C at P and M_P denotes the maximal ideal of this local ring.

Proposition 1.1. *Let C be a curve and $P \in C$ a smooth point. Then $\mathcal{O}_{C,P}$ is a discrete valuation ring.*

The proof of this proposition uses some properties of regular local rings along with Nakayama's lemma. You are strongly encouraged to work out the details.

Because of the proposition, we get the following natural definition.

Definition 1.2. Let C be a curve and $P \in C$ a smooth point. The (normalized) valuation $\mathrm{ord}_P : \mathcal{O}_{C,P} \to \{0, 1, \ldots\} \cup \{\infty\}$ is given by:

$$\mathrm{ord}_P(f) := \sup\{d \geq 0 : f \in M_P^d\}.$$

Using the natural formula $\mathrm{ord}_P(f/g) = \mathrm{ord}_P(f) - \mathrm{ord}_P(g)$, we can extend this valuation to the fraction field of $\mathcal{O}_{C,P}$, which is nothing but $\bar{K}(C)$. So in particular, $\bar{K}(C)$ is equipped with a valuation for every smooth point $P \in C$,

$$\mathrm{ord}_P : \bar{K}(C) \to \mathbb{Z} \cup \{\infty\}.$$

A *uniformizer for C at P* is any function $t \in \bar{K}(C)$ with $\mathrm{ord}_P(t) = 1$, i.e. t is a generator for the maximal ideal M_P.

If C is defined over K and $P \in C(K)$ is smooth, then a uniformizer at P may be chosen in $K(C) \subset \bar{K}(C)$.

Definition 1.3. Let C and P be as in the previous definition, and let $f \in \bar{K}(C)$. Almost tautologically, the order of f at P is $\mathrm{ord}_P(f)$. If this is greater than 0, then f has a zero at P, and if it is less than 0, then f has a pole at P. In this case we write $f(P) = \infty$.

The following proposition should remind you of a similar theorem from complex analysis. And indeed the idea is not very different.

Proposition 1.4. *Let C be a smooth curve and $f \in \bar{K}(C)$ a nonzero element. Then f has finitely many poles and finitely many zeroes in C. Further, if f has no poles, then $f \in \bar{K}$.*

Choice of a uniformizer allows us to say that $K(C)$ is essentially like the field of rational functions in one variable for any curve.

Proposition 1.5. *Let C/K be a curve, and let $t \in K(C)$ be a uniformizer at some nonsingular point $P \in C(K)$. Then $K(C)$ is a finite separable extension of $K(t)$.*

It's not difficult to see that $K(C)$ is a finite extension of $K(t)$, since it is finitely generated over K with transcendence degree 1. The separable part is slightly annoying when characteristic of K is $p > 0$. We leave the details to be checked.

We looked at maps of varieties earlier. Now we specialise to maps of curves which are even more special. In particular, they have a lot more structure.

2. Morphisms between curves

Morphisms of curves are quite rigid in general. In particular, for smooth curves, any rational map is in fact a morphism.

Proposition 2.1. *Let C be a curve and let $V \subset \mathbb{P}^n$ be a variety. Let $P \in C$ be a smooth point and let $\phi : C \dashrightarrow V$ be a rational map. Then ϕ is regular at P. If C is smooth, then ϕ is a morphism.*

Proof. Choose a representation $\phi = [f_0 : \cdots : f_n]$, where $f_i \in \bar{K}(C)$, with the f_i not all zero. Let $r = \min_{0\leq i\leq n} v_P(f_i)$. Since the f_i are not all zero, the integer r is finite. Choose a uniformizer t at P and let $g = t^{-r} \in \bar{K}(C)^\times$. Then $\mathrm{ord}_P(gf_i) = \mathrm{ord}_P(f_i) - r \geq 0$ for every i. Hence $gf_i \in \mathcal{O}_{C,P}$ for every i. Moreover, by the definition of r, there is at least one index j such that $\mathrm{ord}_P(gf_j) = 0$. Thus gf_j is a unit in $\mathcal{O}_{C,P}$. It follows that $[(gf_0)(P) : \cdots : (gf_n)(P)]$ is a well-defined point of $\mathbb{P}^n$.

The following expression makes sense for every homogeneous polynomial $F \in I(V)$ of degree d:

$$F\big((gf_0), \ldots, (gf_n)\big) = g^d F(f_0, \ldots, f_n) = 0.$$

Since each gf_i is regular at P, we can evaluate this at P. Hence, $[(gf_0)(P) : \cdots : (gf_n)(P)] \in V$. Hence, ϕ is regular at P. □

You should think of the previous proposition as 'functions on curves have removable singularities'.

In fact, morphisms between two curves are even more restricted.

Theorem 2.2. *let $\phi : C_1 \to C_2$ be a morphism of curves. Then ϕ is either surjective or a constant morphism. That is, a nonconstant morphism of curves must be surjective.*

Proof. Since C_1 is projective, its image $\phi(C_1)$ is a closed subset of C_2. Moreover, since C_1 is irreducible, its image $\phi(C_1)$ is irreducible. (This is so because every morphism of projective varieties is continuous for the zariski topology - something that we have not discussed too much.)

Every proper closed subset of the irreducible curve C_2 has dimension zero, and is therefore finite. Since $\phi(C_1)$ is also irreducible, if it is a proper closed subset of C_2, it must consist of a single point. In that case ϕ is constant. Otherwise, $\phi(C_1) = C_2$, and hence ϕ is surjective. □

Remark 2.3. Let C be a smooth curve and $f \in K(C)$. Then f defines a rational map $f : C \to \mathbb{P}^1$ by the formula $[f(P) : 1]$. By the previous proposition, this is a morphism. Indeed the only place where f may have difficulty being defined is where it has poles, and at such points we simply extend the above definition by setting $f(P) = [1, 0]$ if f has a pole at P. Conversely, any rational map $\phi : C \to \mathbb{P}^1$ is given by a tuple $[f : g]$ where if $g \neq 0$, ϕ corresponds to the function $f/g \in K(C)$. If $g = 0$, then ϕ is the constant map that maps every point to $[1 : 0]$. Denoting this map by the symbol ∞, we get a one-to-one correspondence:

$$K(C) \cup \{\infty\} \longleftrightarrow \{f : C \to \mathbb{P}^1 \text{ defined over } K\}.$$

Given this correspondence between curves and their function fields, the next result should not come as too much of a surprise. We will not give its proof, but use it wholeheartedly during the course.

We will assume every curve occurring from now in this lecture is smooth. Let C_1/K and C_2/K be curves. Let $\phi : C_1 \to C_2$ be a nonconstant morphism defined over K. Then note that composition with ϕ induces an injection of function fields:

$$\phi^* : K(C_2) \to K(C_1), \qquad \phi^* f = f \circ \phi.$$

Theorem 2.4. *Let C_1/K and C_2/K be curves and ϕ as above.*

- *$K(C_1)$ is a finite extension of $\phi^*(K(C_2))$.*
- *Let $\iota : K(C_2) \to K(C_1)$ be an injection of fields that fixes K. Then there exists a unique nonconstant map $\phi : C_1 \to C_2$ defined over K such that $\phi^* = \iota$.*
- *Let $\mathbb{K} \subset K(C_1)$ be a subextension of finite index that contains K. Then there exist a smooth curve C'/K, unique up to K-isomorphism, and a nonconstant map $\phi : C_1 \to C'$ defined over K such that $\phi^*(K(C')) = \mathbb{K}$.*

This gives an important definition for maps between curves.

Definition 2.5. Let $\phi : C_1 \to C_2$ be a morphism of curves defined over K. If ϕ is a constant map, we define the *degree* of ϕ to be equal to 0. Otherwise we say ϕ is a *finite* morphism and define its degree to be

$$\deg \phi = [K(C_1) : \phi^*(K(C_2)].$$

We say that ϕ is *separable*, *inseparable*, or *purely inseparable* if the corresponding field extension has the corresponding property, and we denote the separable and inseparable degrees of the field extension by $\deg_s \phi$ and $\deg_i \phi$ respectively.

We give another important definition.

Definition 2.6. Let $\phi : C_1 \to C_2$ be a nonconstant morphism of curves defined over K. We define:

$$\phi_* : K(C_1) \to K(C_2), \qquad \phi_* = (\phi^*)^{-1} \circ Nm_{K(C_1)/\phi^*(K(C_2))}$$

where $Nm_{E/F}$ denotes the norm map for the field extension E/F.

We end with the following corollary:

Corollary 2.7. *Let C_1 and C_2 be smooth curves over K and let $\phi : C_1 \to C_2$ be a morphism of degree 1 defined over K. Then ϕ is an isomorphism.*

Remark 2.8. All this shows that there is an anti-equivalence of categories $\mathcal{C}$ and $\mathcal{F}$; where $\mathcal{C}$ is the category of smooth curves defined over K with morphisms given by nonconstant rational maps defined over K, and $\mathcal{F}$ is the category of finitely generated field extensions $\mathbb{K}/K$ of transcendence degree 1 satisfying $\mathbb{K} \cap \bar{K} = K$ with morphisms given by field injections fixing K.